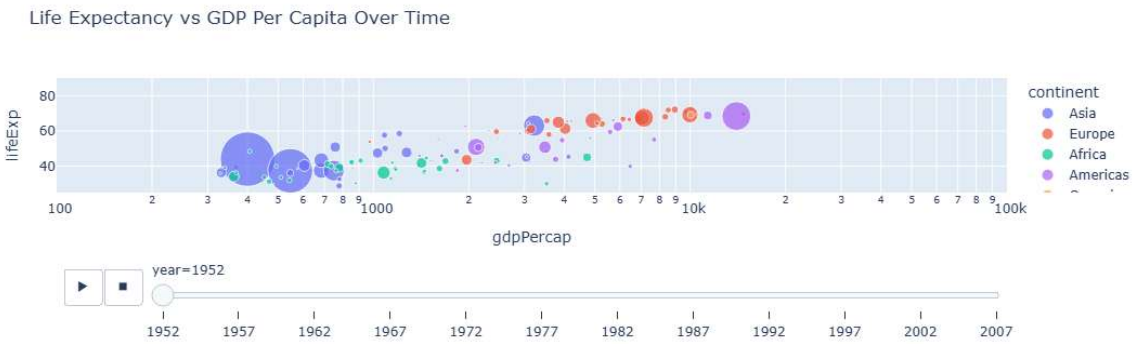
In [3]:
```
import pandas as pd
import plotly.express as px

# 1. Load a sample dataset (Gapminder tracks country data over time)
df = px.data.gapminder()

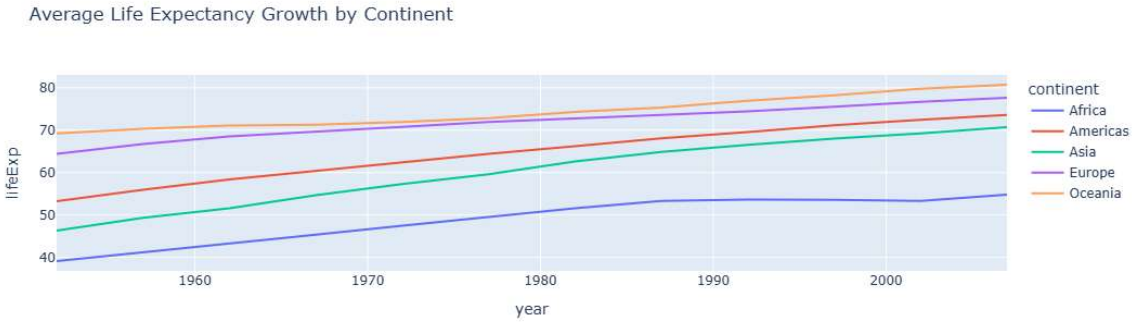
# 2. Create an Interactive Scatter Plot
fig = px.scatter(df, x="gdpPercap", y="lifeExp",
                 animation_frame="year",
                 animation_group="country",
                 size="pop", color="continent",
                 hover_name="country",
                 log_x=True, size_max=55,
                 range_x=[100,100000], range_y=[25,90],
                 title="Life Expectancy vs GDP Per Capita Over Time")

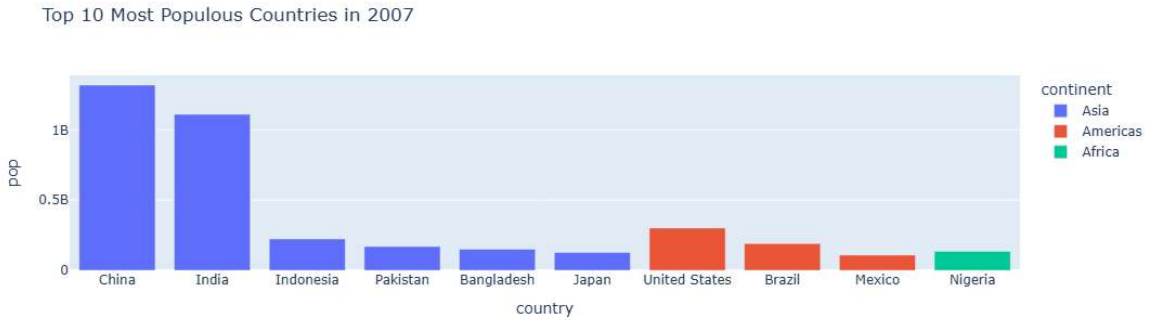
# 3. Display the dashboard
fig.show()
```



In [4]:
```
# 1. Line Chart: Life Expectancy Trend by Continent
df_continent = df.groupby(['year', 'continent'])['lifeExp'].mean().reset_index()
fig_line = px.line(df_continent, x="year", y="lifeExp", color="continent",
                   title="Average Life Expectancy Growth by Continent")
fig_line.show()

# 2. Bar Chart: Population in 2007 (The most recent year in the data)
df_2007 = df[df['year'] == 2007]
fig_bar = px.bar(df_2007.sort_values('pop', ascending=False).head(10),
                 x='country', y='pop', color='continent',
                 title="Top 10 Most Populous Countries in 2007")
fig_bar.show()
```





Task 3: Executive Summary & Insights

1. Economic Impact on Health

The interactive scatter plot reveals a clear logarithmic relationship between **GDP per Capita** and **Life Expectancy**. Initial economic growth provides massive jumps in health outcomes, while "developed" nations see diminishing returns, suggesting health investments move from "survival" to "quality of life" as wealth increases.

2. Global Progress Trends

The line chart demonstrates a consistent upward trend across all continents. Notably, **Asia** shows the steepest growth curve, nearly closing the gap with the Americas and Europe over the 50-year period.

3. Population Concentration

The bar chart highlights that global population is heavily concentrated in the **Asian continent** (specifically China and India). For any global health or economic policy, these two nations represent the highest impact areas for resource allocation.

Deliverable Status: Fully functional interactive dashboard completed.

In [ ]: